Valve Design Project

	2.40	1.73	1.48	93.8	
(A)	5.00	1.00	2.45		45.1
(B)	2.322	1.857	1.427	94.7	93.8
(C)	2.58	1.73	1.36	95	94.2
(D)	2.40	1.73	1.75		94.3
(E)	2.82	1.61	1.48		93.9

PROJECT 4 : Chemical Equilibrium

$$\min \quad f(x) = \sum_{j=1}^{n=10} x_j \left(c_j + \ln \frac{x_j}{s} \right),$$

$$\text{s.t.} \quad x_1 + 2x_2 + 2x_3 + x_6 + x_{10} = 4$$
$$x_4 + 2x_5 + x_6 + x_7 = 2$$
$$x_3 + x_7 + x_8 + 2x_9 + x_{10} = 2$$

$x > 0$

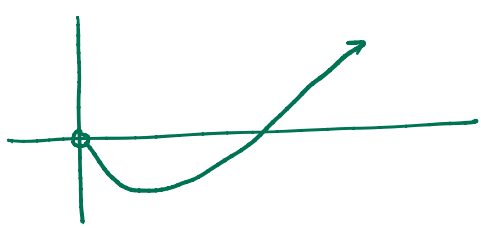
$x \in \mathbb{R}^{10}$

(j)	component	c_j
1	H	-6.089
2	H_2	-17.164
3	H_2O	-34.054
4	N	-5.914
5	N_2	-24.721
6	NH	-14.986
7	NO	-24.100
8	O	-10.708
9	O_2	-26.662
10	OH	-22.179

$$s = \sum_{j=1}^{10} x_j$$

$$N_2H_4 + O_2 \rightleftharpoons x_1 H + x_2 H_2 + x_3 H_2O + \cdots + x_{10} OH$$

x_j = (# moles of component j in the eq. mixture)

The key difficulty with the objective/constraints is that $\ln x_k/s$ is not defined whenever $x_k/s \leq 0$. This can occur when $x_k \leq 0$ and $s > 0$ or when $x_k > 0$ and $s_k < 0$. It is also not defined when $s = 0$. We cannot simply enforce $x_k/s > 0$ directly as infeasible point algorithms need to evaluate such points. We need to modify the constraints $x, s > 0$ to gracefully handle $x \leq 0$ and $s \leq 0$.

$x_j \ln x_j/s$